

UNIFIED GEOMETRIC-TOPOLOGICAL SUBSTRATE

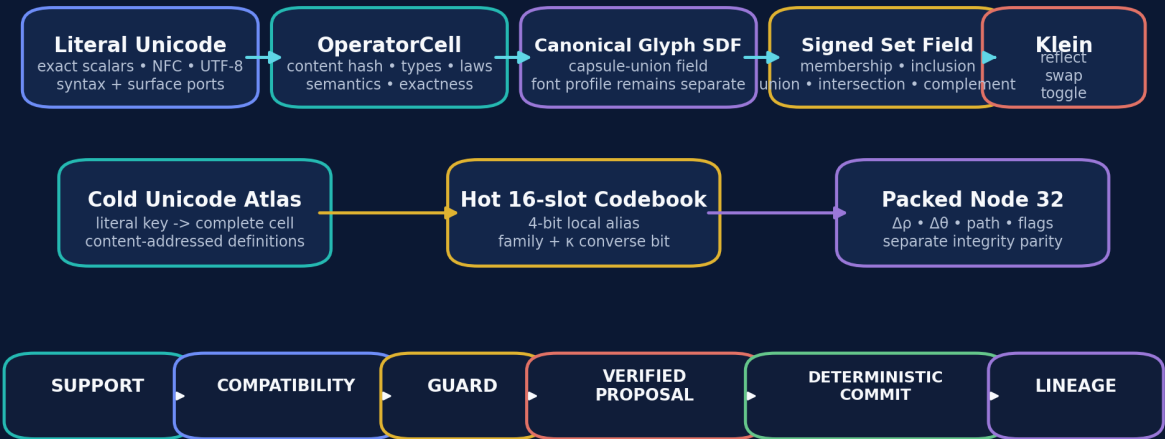
UGTS FOUNDATION 5.0

UNICODE SET-FIELD AND KLEIN-CONVERSE ATLAS

Content-addressed operator cells, signed set fields, reflective operand converse and packed log-polar execution

UGTS Foundation 5.0 Architecture

Unicode literal, field semantics, converse topology and packed execution remain inside one audited authority chain



Pure operator evaluation proposes evidence; only the retained UGTS chain may change authoritative state.

Canonical identity	ugts.foundation.unicode-set-field-klein-converse@5.0.0
Legacy discovery alias	UGTS-KC 5.0
Component type	Foundation
Version / date	5.0.0 / 29 August 2026
Codename	USF-KCA
Mechanism delta	USF001-USF064; legacy continuity M886-M949
Reference validation	83 Python tests PASS; 3 schema instances with zero errors; C++20 scalar test PASS
Prepared for	Tom Klootwijk

ATTRIBUTION AND EVIDENCE BOUNDARY. Requester-supplied attribution; technical artifact, not legal proof. Exact-SDF universality, finite mathematical completeness and physical SIMD performance are not claimed.

Executive Summary

FORMAL DECISION

Release UGTS Foundation 5.0 as a component-scoped major version. The literal Unicode operator becomes the cold-atlas key for a content-addressed *OperatorCell* that co-addresses exact Unicode spelling, typed syntax, semantic evaluator, canonical glyph SDF, signed set-field action, converse relation, Klein reflection rule, exactness capability, provenance and content hash. Geometry and semantics remain distinct typed faces; they are not collapsed into one notion of equality.

VERIFIED RELEASE RESULT

The accompanying package supplies 20 hash-verified *OperatorCells*; a 16-slot set-relation hot codebook with 14 assigned converse slots and two reserved slots; a corrected Packed Set-Field Node 32 codec; binary stream framing with CRC; a dependency-free Python reference runtime; portable C++20 scalar packing; an optional AVX2 extraction profile; JSON Schemas; a 64-mechanism delta; a 24-row claims ledger; runnable examples; a wheel; and captured validation evidence. The release test suite contains 83 passing Python tests. Three JSON instances validate with zero errors, and the native scalar codec test passes.

MAJOR-VERSION CONCLUSION

A Unicode operator is no longer merely text and no longer merely an icon. It is a content-addressed, executable geometric rule cell whose literal codepoint resolves both its canonical shape and its typed operation. A Klein-converse parity flip mirrors the shape, reverses the typed ports and selects the converse Unicode spelling while preserving the mathematical truth value.

EVIDENCE BOUNDARY

The immediate source document correctly identifies $\exists$ as the converse spelling of ϵ , proposes geometric operator primitives, log-polar fixed-point packing, a 32-bit node and SIMD decoding. It also contains packing errors, overloaded parity, incomplete geometry, and unsupported universal compression, speed, randomness and physical claims. Version 5.0 does not silently clean the source into a fictional proof. Every major claim is admitted, corrected, bounded or rejected in the machine-readable claims ledger. [S0]

Contents

Executive Summary	i
1 Major Release Identity and Scope	1
1.1 Why this is a major version	1
1.2 Component-scoped identity	1
1.3 What 5.0 adds	2
1.4 What remains unchanged	2
2 Source Basis and Evidence Discipline	2

2.1	Registered sources	2
2.2	Immediate source dissection	3
2.3	Inherited commitments	3
2.4	Evidence classes	4
3	Formal 5.0 Architecture	4
3.1	Upgraded world object	4
3.2	Canonical processing path	4
3.3	Single source of operator truth	5
4	The Unicode OperatorCell	5
4.1	Formal record	5
4.2	Literal identity	5
4.3	Typed ports and reversed operand order	6
4.4	Three faces, one address	6
4.5	Content addressing	6
5	Set Theory as Signed Set Fields	6
5.1	General metric definition	6
5.2	Finite and symbolic fallback	7
5.3	Set algebra as field algebra	7
5.4	Relation kernels	8
5.5	Capability-aware certificates	8
6	Unicode Converse Families	9
6.1	Family plus κ	9
6.2	Semantic converse law	9
6.3	Algebra operators outside the hot pair codebook	9
7	The Klein-Converse Involution	9
7.1	Local transform	9
7.2	Local involution and lineage	9
7.3	Reflective quotient	10
7.4	Discrete angle maps	10
8	Canonical Glyph SDFs	10
8.1	Canonical profile	10
8.2	Mirror law	11
8.3	Font-exact mode	11
9	Cold Atlas, Hot Codebook and LUT	12
9.1	Cold atlas	12
9.2	Hot codebook	12

9.3 Literal-to-LUT path	13
10 Packed Log-Polar Node 32	13
10.1 Bit layout	14
10.2 Packing formula	14
10.3 Two parities, two roles	14
10.4 Mandatory stream header	14
10.5 Quantization	14
10.6 Bounded grammar path	15
11 Scalar and SIMD Execution Profiles	15
11.1 Scalar oracle	15
11.2 AVX2 eight-lane extraction	15
11.3 ARM NEON profile	15
11.4 Branch strategy	15
11.5 Benchmark boundary	16
12 Exactness, Proof and Certification	16
12.1 Point query versus global relation	16
12.2 Certificate object	16
12.3 Finite exact proof	16
12.4 Continuous proof	16
12.5 Integrity ladder	17
13 Integration with UGTS State Authority	17
13.1 Pure evaluation is upstream	17
13.2 Canonical handoff	17
13.3 Example: membership-triggered transition	17
13.4 Example: reflective route transition	17
14 Reference Implementation and Validation	17
14.1 Package map	18
14.2 Validation result	18
14.3 What the tests establish	19
14.4 What the tests do not establish	19
14.5 Reproduction	19
15 Claims Ledger Summary	19
16 Acceptance, Kill Criteria and Final Definition	20
16.1 Acceptance gates	20
16.2 Kill criteria	20
16.3 Final upgraded definition	21

A	Complete Shipped Unicode Operator Atlas	22
B	Hot Codebook Set-Core-16	22
C	Complete Mechanism Delta USF001-USF064	22
D	Complete Claims Ledger	24
E	Source Hashes and Package Reproduction	25
E.1	Full source hash register	25
E.2	Artifact identity	25
E.3	Canonical filename	26
E.4	Release status	26

1 Major Release Identity and Scope

1.1 Why this is a major version

Version 5.0 changes the unit of formal operator identity. Earlier releases already established content-addressed definitions, representation profiles, log-polar packing, explicit reflective topology, general typed operators and field-capability boundaries. This release unifies those commitments in a new foundation component whose central object is the Unicode-addressed executable operator cell.

The major change is not a larger list of symbols. It is a stronger invariant:

The exact Unicode literal is a stable lookup key into one record that carries syntax, typed semantics, canonical geometry, field behavior, converse topology and execution metadata.

A change in font outline does not silently change set membership. A change in operand spelling from $x \in A$ to $A \ni x$ does not create a second mathematical relation. A packed four-bit slot does not replace the global Unicode identity. These distinctions now belong to the formal release schema rather than explanatory prose.

1.2 Component-scoped identity

The versioning charter deprecates a single global ladder because compute profiles, foundations, runtimes, formal packs, applications and platform builds are not globally comparable. Version 5.0 therefore uses the canonical identity

`ugts.foundation.unicode-set-field-klein-converse@5.0.0`.

The legacy alias **UGTS-KC 5.0** remains for discovery. The parent graph records exactly which releases contribute architecture; it does not claim that this foundation supersedes unrelated game, scanner, solver or device branches. [S7]

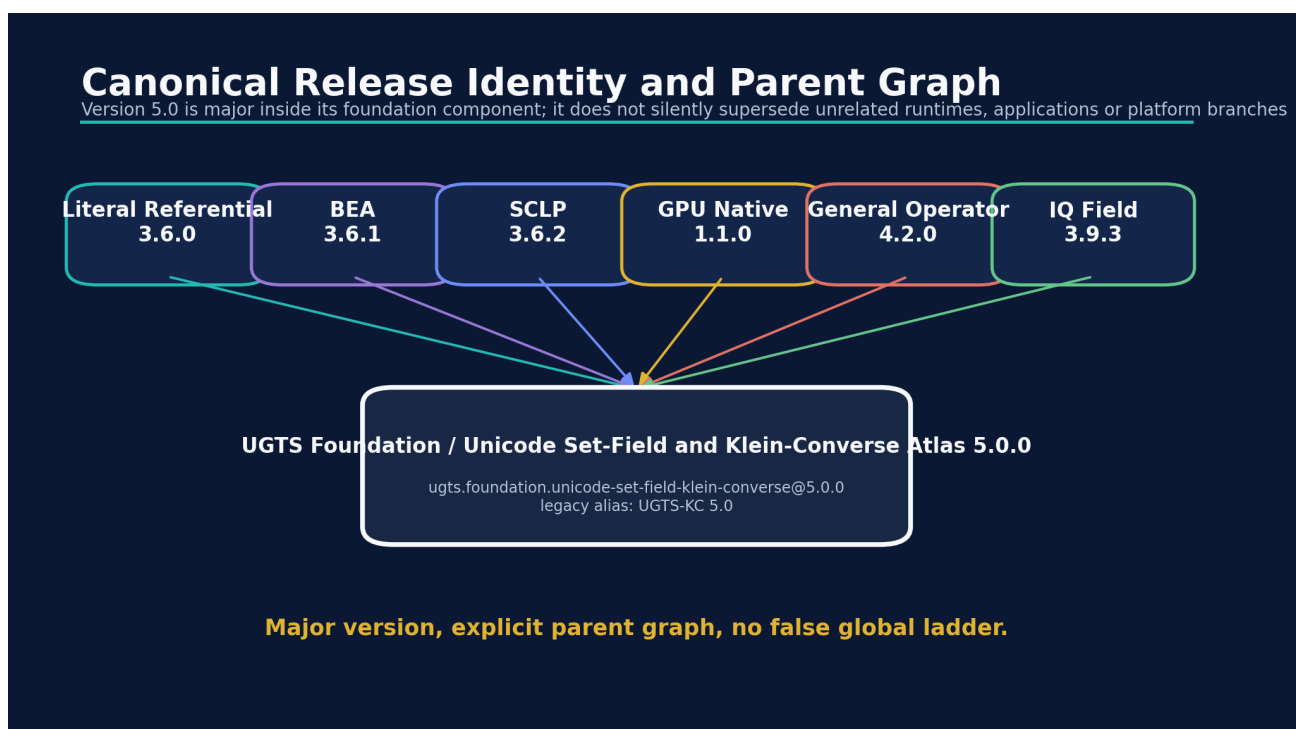


Figure 1: Version 5.0 is major within its foundation component. Typed parent edges replace a false global sequence.

1.3 What 5.0 adds

- A content-addressed **Unicode OperatorCell** schema.
- A six-level signed set-field capability ladder.
- Canonical set algebra over sign-compatible fields.
- Direct/converse Unicode relation families with explicit surface-to-canonical port maps.
- A reflective **Klein-converse involution**: operator converse, operand swap, chart reflection, orientation flip and κ -toggle.
- A canonical capsule-union glyph SDF profile plus a font-exact profile boundary.
- A cold Unicode atlas and a hash-bound hot 16-slot codebook.
- A corrected 32-bit packed node and mandatory stream header.
- Scalar, AVX2 and NEON execution contracts with a scalar oracle.
- Exactness, quantization, integrity and event-margin certificates.
- 64 new mechanisms and a complete claims ledger.

1.4 What remains unchanged

The canonical UGTS authority order is preserved:

support → compatibility → guard → verified proposal → deterministic commit → lineage.

Pure operator evaluation may produce evidence or a proposal. It cannot mutate authoritative state, topology, ownership or lineage by itself. Projection, glyph rendering, SIMD layout and packed storage remain replaceable implementations around the authority chain. [\[S4\]](#) [\[S5\]](#)

2 Source Basis and Evidence Discipline

2.1 Registered sources

ID	Source	SHA-256 prefix	Use in 5.0
S0	mandible-broad-set-math-operator-CE-log-o-like-simd256.pdf	6149e3d398c2...	Immediate Unicode/SDF/log-polar/SIMD source motif
S1	UGTS_KC_3_6_Tom_Klootwijk.pdf	e21433cd8378...	Literal content-addressed definitions and operation order
S2	UGTS_KC_3_6_1_BEA_Tom_Klootwijk.pdf	ac857c9c01f5...	Representation profile, glyph SDF and typed semantic evaluator separation
S3	UGTS_KC_3_6_2_SCLP_Tom_Klootwijk.pdf	6987490a0a7b...	Log-polar chart, finite packing, half-turn/Klein distinction
S4	Unified_Geometric_Topological_Substrate_GPU_Native_Addendum.pdf	3a6b6c57118c...	Packed-state, one-bit and hardware evidence discipline
S5	UGTS_KC_4_2_General_Operator_Order_Addendum(1).pdf	413ef9568330...	General typed operator record and explicit order calculus
S6	UGTS_KC_3_9_3_IQ_Field_Substrate_Tom_Klootwijk(1).pdf	c0ae58652436...	Field exactness/capability and sign-preserving CSG boundary

ID	Source	SHA-256 prefix	Use in 5.0
S7	UGTS_Versioning_Charter_Phase_1_Foundation_Chronology(1).pdf	1b64495ed32a...	Component-scoped canonical identity and filename rule

2.2 Immediate source dissection

The immediate source contributes five technically useful motifs:

1. $\in$ and $\ni$ are converse surface spellings of one membership relation.
2. Unicode operator shapes can be described by bounded geometric primitives.
3. local geometry can be addressed in log-polar coordinates and packed into fixed-width records.
4. a one-bit selector can choose between declared orientations or relation directions.
5. SIMD extraction can decode several packed words in parallel.

It also requires explicit correction:

- The source's example LUT selects codepoints by quadrant rather than evaluating the advertised glyph SDFs.
- One packing expression overlaps angle and codepoint bits.
- The proposed 32-bit decoder extracts the parity bit but does not verify payload parity.
- The active bit is omitted by one unpacker.
- A Boolean value of 0 or 1 is not a valid full byte/lane mask for `blendv`; it must be expanded.
- A deterministic parity bit is not true randomness, anti-aliasing proof, complete state or cryptographic entropy.
- A half-turn rotation is not an orientation-reversing Klein gluing.
- Universal 16 KB, constant-time, cycle-count and collision-free-path claims are not established.

2.3 Inherited commitments

Literal referential 3.6.0 makes definitions first-class, content-addressed substrate data with resolvable dependencies and explicit evaluation order. [\[S1\]](#)

BEA 3.6.1 defines a representation profile containing normalization, glyph renderer, byte encoder and typed semantic evaluator. It insists that glyph SDF, semantic residual and encoding alignment remain separate spaces. Version 5.0 adopts that separation but lets one Unicode cell co-address all three. [\[S2\]](#)

SCLP 3.6.2 supplies the log-polar chart and the decisive distinction between a source half-turn bundle and a genuine reflective Klein profile:

$$(\rho, \theta, \varphi, o) \mapsto (\rho', \pi - \theta, -\varphi, -o).$$

The angular derivative is -1 , so the base gluing reverses orientation. [\[S3\]](#)

GPU Native 1.1 supplies the compact-state, narrow-one-bit, compression and hardware-evidence boundaries. [\[S4\]](#)

General Operator and Order 4.2 supplies the typed OperatorSpec, explicit order profiles, exactness/effect/capability metadata and the rule that pure evaluation cannot bypass event authority. [S5]

IQ Field 3.9.3 supplies the capability distinction between exact distance, conservative bounds and sign-correct residuals, including the warning that min/max CSG generally preserves sign but not exact Euclidean distance. [S6]

2.4 Evidence classes

Class	Treatment in 5.0
Source-derived motif	Unicode converse, geometric glyph proposal, log-polar packing, branchless selector and SIMD sketch are registered as source motifs.
Engineering-derived	OperatorCell schema, signed set-field calculus, canonical capsule glyphs, Klein-converse transform, codebook and codecs are new formalization.
Measured package evidence	Tests, hashes, schema results, binary round trips, wheel and native scalar outputs are captured in validation/.
Bounded hypothesis	AVX2/NEON speed, cache behavior and target-device advantage remain benchmark questions.
Rejected claim	Universality, infinite detail, true randomness, automatic collision freedom and physical spacetime assertions are excluded.

3 Formal 5.0 Architecture

3.1 Upgraded world object

FORMAL DEFINITION

Let the version-5 foundation object be

$$W_5 = W_{\text{retained}} + (\mathcal{U}, \mathcal{O}, \Phi, \Gamma, \mathcal{K}, \mathcal{C}, \mathcal{P}, \text{Cert}),$$

where $\mathcal{U}$ is the Unicode literal domain; $\mathcal{O}$ the content-addressed OperatorCell atlas; Φ the signed set-field calculus; Γ canonical glyph fields; $\mathcal{K}$ the Klein-converse transform; $\mathcal{C}$ hot codebooks; $\mathcal{P}$ packed execution records; and Cert exactness, integrity, quantization and event-margin certificates.

3.2 Canonical processing path

```

literal Unicode sequence
-> normalize under declared profile
-> resolve OperatorCell + content hash
-> map surface operands to canonical typed ports
-> evaluate semantic relation / set-field transform
-> evaluate optional canonical glyph SDF
-> apply optional Klein-converse chart transition
-> resolve hot codebook slot
-> pack or decode bounded execution record
-> issue capability + error + integrity certificate
-> support -> compatibility -> guard -> verified proposal
-> deterministic commit -> lineage / replay

```

-> optional render, storage, SIMD or device adapter

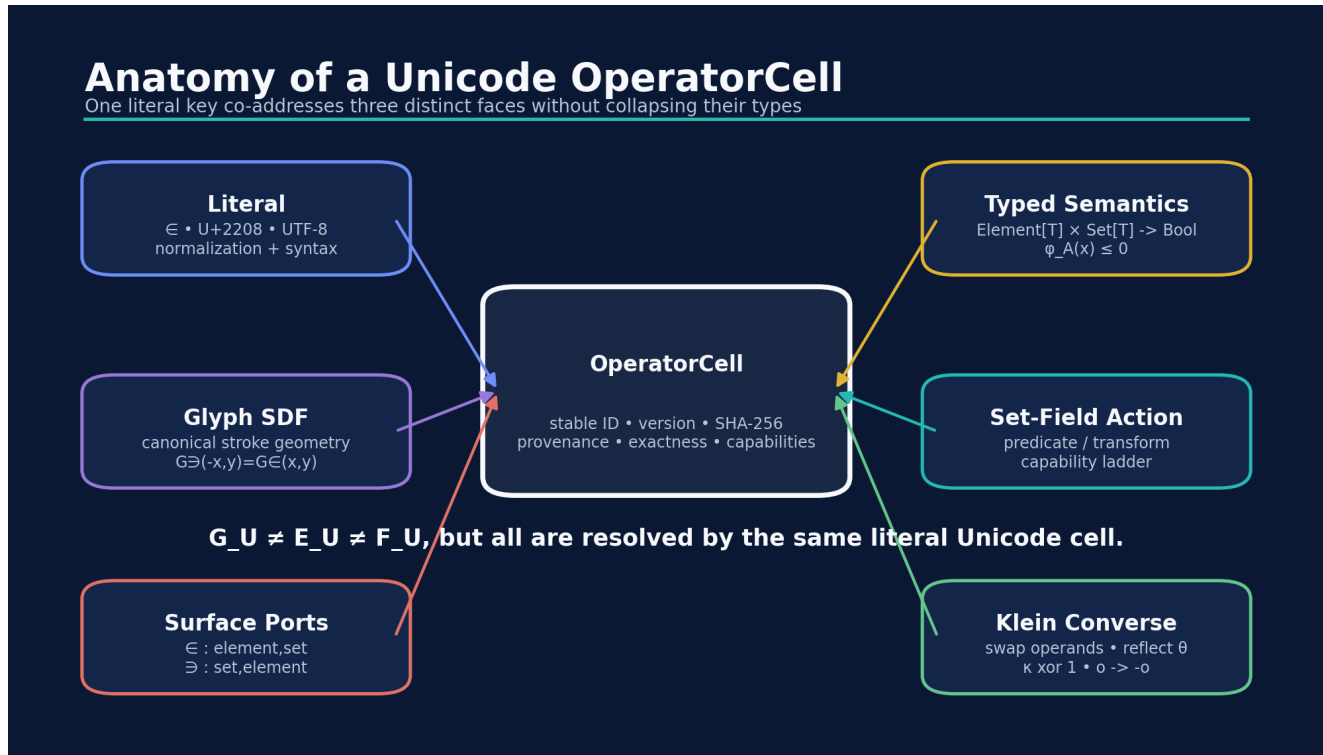


Figure 2: The literal, glyph, semantics and field action are distinct but co-addressed.

3.3 Single source of operator truth

The same OperatorCell records feed schema validation, Unicode lookup, relation evaluation, glyph generation, converse mapping, hot-codebook construction, content hashing, report tables and tests. A prose-only operator meaning cannot override the machine-readable record.

4 The Unicode OperatorCell

4.1 Formal record

FORMAL DEFINITION

For exact Unicode scalar sequence U , define

$$\mathcal{O}_U = (U, \ell_U, G_U, E_U, F_U, C_U, K_U, P_U, h_U),$$

where ℓ_U is the literal spelling; G_U canonical glyph SDF; E_U typed semantic evaluator; F_U set-field action; C_U typed ports and surface order; K_U converse/topology rule; P_U representation and exactness profile; and h_U the canonical content hash.

The record stored in `spec/operator_atlas.json` additionally carries a stable ID, version, provenance, Unicode names, UTF-8 bytes, laws, capabilities and explicit non-claims.

4.2 Literal identity

The cold-atlas key is the exact NFC-normalized scalar sequence, not a font outline and not a short local opcode. For $\in$:

Literal	€
Unicode scalar	U+2208
Unicode name	ELEMENT OF
UTF-8 bytes	e2 88 88
Stable operator ID	set.membership.element-of

Compatibility normalization is deliberately not applied because it can collapse presentation-distinct mathematical symbols. A profile may register an alias, but the alias must point to a stable operator identity rather than silently replacing the literal.

4.3 Typed ports and reversed operand order

For membership, the canonical semantic ports are

[element, container].

The two surface forms are

$$x \in A \quad \text{and} \quad A \ni x.$$

The direct syntax map is $[0, 1]$; the converse syntax map is $[1, 0]$. Both call one membership kernel after port normalization. This makes operand reversal executable data rather than a parser special case.

4.4 Three faces, one address

The release enforces

$$G_U \neq E_U \neq F_U,$$

yet resolves all three through the same literal cell. A glyph sample does not decide membership. A semantic truth value does not imply zero geometric glyph distance. A set-field transform does not establish a font's outline. The co-addressing is a referential convenience with typed boundaries, not a claim of cross-domain identity. [\[S2\]](#)

4.5 Content addressing

Each cell hash is computed from canonical UTF-8 JSON with sorted keys, NFC strings and no non-finite numbers, excluding only the hash field itself. The atlas hash commits to all cells and atlas metadata. A codebook commits to the atlas hash and each occupied operator-cell hash.

BOUNDARY

A SHA-256 content address identifies bytes under the canonicalization policy. It is not legal authorship, ownership, priority, semantic truth or collision-free identity across every possible future policy.

5 Set Theory as Signed Set Fields

5.1 General metric definition

Let X be a declared universe with metric or graph distance d , and $A \subseteq X$. Define

$$\phi_A(x) = d(x, A) - d(x, X \setminus A).$$

Then, under the declared boundary policy,

$$\phi_A(x) < 0 \text{ inside,} \quad \phi_A(x) = 0 \text{ on the boundary,} \quad \phi_A(x) > 0 \text{ outside.}$$

Membership becomes

$$x \in A \iff \phi_A(x) \leq 0.$$

The definition is exact only when the metric and the representations of A and its complement are exact. Sampled point sets generally produce an approximation or bound rather than an exact continuous SDF.

5.2 Finite and symbolic fallback

When no useful metric exists, a signed characteristic field is sufficient:

$$\chi_A(x) = \begin{cases} -1, & x \in A, \\ +1, & x \notin A. \end{cases}$$

The reference runtime implements `FiniteSetField` over a declared ordered finite universe. Its capability is `signed_membership_field`, not `exact_sdf`.



Figure 3: Every field carries the strongest capability actually established.

5.3 Set algebra as field algebra

With the negative-inside convention:

$$\begin{aligned} \phi_{A \cup B} &= \min(\phi_A, \phi_B), \\ \phi_{A \cap B} &= \max(\phi_A, \phi_B), \\ \phi_{X \setminus A} &= -\phi_A, \\ \phi_{A \setminus B} &= \max(\phi_A, -\phi_B), \\ \phi_{A \triangle B} &= \max(\min(\phi_A, \phi_B), -\max(\phi_A, \phi_B)). \end{aligned}$$

These formulas preserve the correct inside/outside sign. They do not automatically preserve exact Euclidean distance after composition. [\[S6\]](#)

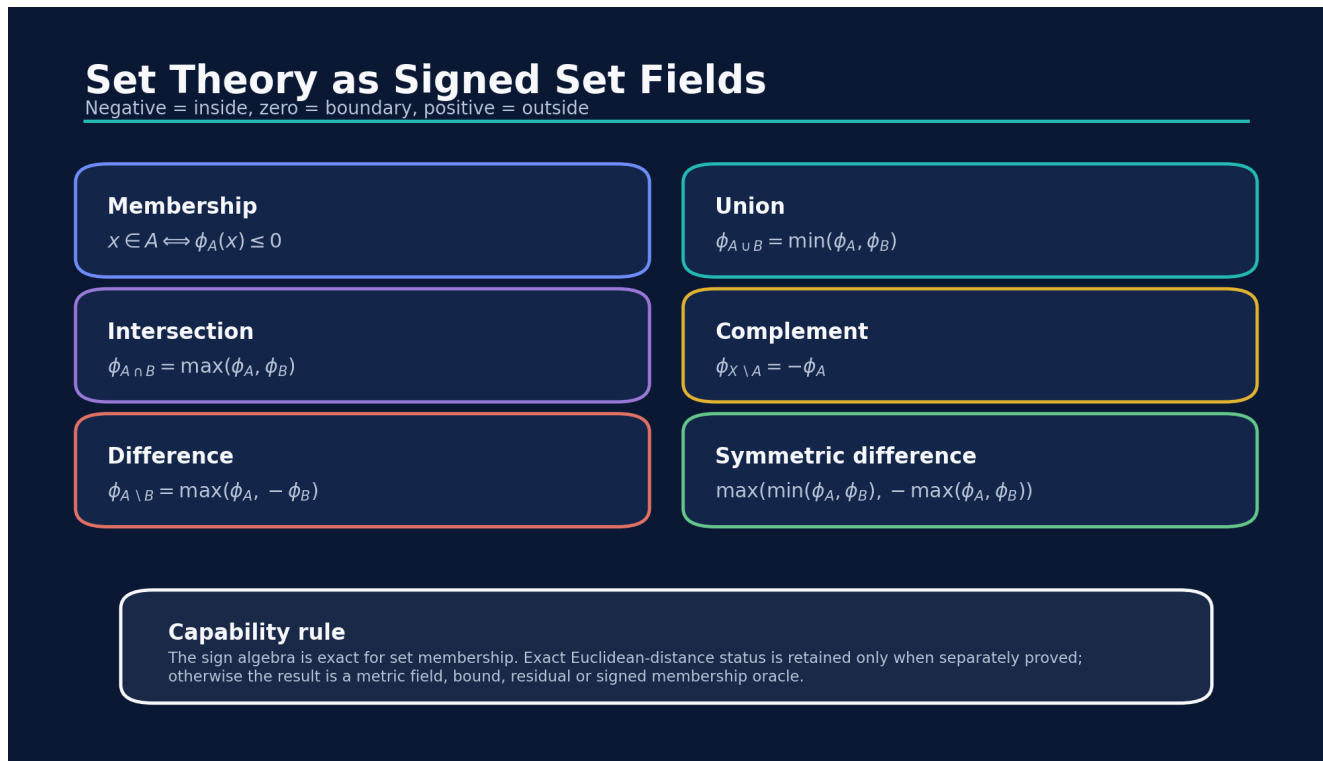


Figure 4: The field layer turns set operations into bounded, branch-friendly scalar operations.

5.4 Relation kernels

The shipped reference subset implements:

$$\begin{aligned}
 x \in A &\iff \phi_A(x) \leq 0, \\
 x \notin A &\iff \phi_A(x) > 0, \\
 A \subseteq B &\iff A \setminus B = \emptyset, \\
 A \subset B &\iff A \subseteq B \wedge A \neq B, \\
 A = B &\iff A \subseteq B \wedge B \subseteq A.
 \end{aligned}$$

On a finite universe these statements may be checked exhaustively. On a continuous universe, a sampled LUT is not a universal emptiness or inclusion proof.

5.5 Capability-aware certificates

A set-field result carries at least:

- universe/profile identity;
- sign convention and boundary inclusion policy;
- capability class;
- metric or oracle identity;
- numeric error or interval;
- proof scope: point query, finite exhaustive domain, sampled domain or continuous certificate;
- explicit non-claims.

6 Unicode Converse Families

6.1 Family plus κ

Each hot relation family uses

$$\text{operator_id} = (\text{family} \ll 1) \mid \kappa, \quad \kappa \in \{0, 1\}.$$

The family selects the canonical semantic relation. $\kappa = 0$ selects the direct spelling; $\kappa = 1$ selects the converse spelling and reversed surface ports.

Family	$\kappa = 0$	$\kappa = 1$	Canonical relation
0	$x \in A$	$A \ni x$	membership
1	$x \notin A$	$A \not\ni x$	nonmembership
2	$A \subset B$	$B \supset A$	proper inclusion
3	$A \subseteq B$	$B \supseteq A$	inclusion or equality
4	$A \not\subset B$	$B \not\supset A$	negated proper inclusion
5	$A \not\subseteq B$	$B \not\supseteq A$	negated inclusion or equality
6	$A \subsetneq B$	$B \supsetneq A$	strict-inclusion glyph variant
7	reserved	reserved	codebook extension gate

6.2 Semantic converse law

For every registered pair $U, U^\smile$,

$$E_{U^\smile}(b, a) = E_U(a, b).$$

This is relation converse, not functional inversion. Typed endpoints swap; multiplicity, provenance and truth are preserved.

6.3 Algebra operators outside the hot pair codebook

The cold atlas also ships $\cup$, $\cap$, $\setminus$, $\mathbb{C}$, $\emptyset$ and $=$. They are addressed by Unicode and carry geometry plus functionality, but they do not consume a direct/converse slot in the shipped set-core-16 codebook. A later codebook may assign them local slots without changing their global identities.

7 The Klein-Converse Involution

7.1 Local transform

Define the local transform

$$\mathcal{K}(U, a, b, \rho, \theta, \kappa, o) = (U^\smile, b, a, \rho, \pi - \theta, \kappa \oplus 1, -o).$$

It acts simultaneously on literal spelling, typed ports, glyph orientation, chart angle and orientation state.

7.2 Local involution and lineage

On the local fiber,

$$\mathcal{K}^2 = \text{id}.$$

After two applications the literal, ports, angle, κ and orientation return. Lineage does *not* erase the two wraps; the winding counter increases by two. Thus the state transform is involutive while history remains auditable.

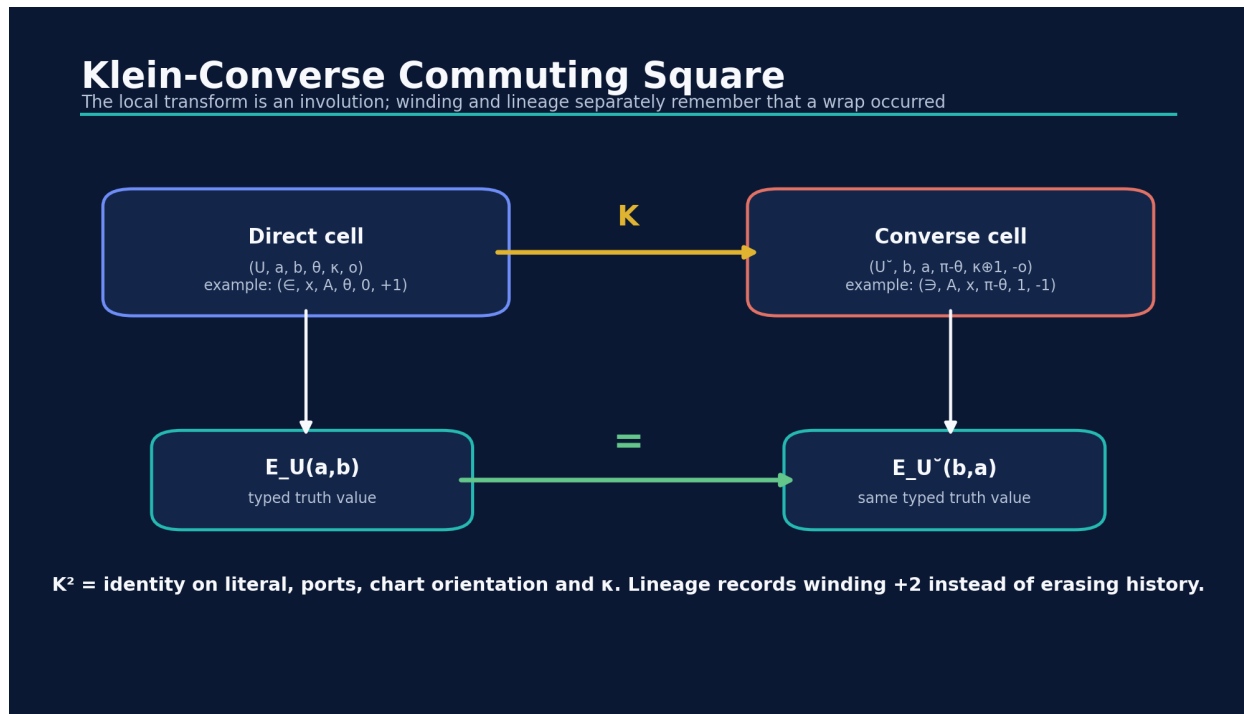


Figure 5: The semantic square commutes while lineage separately records the wrap.

7.3 Reflective quotient

A genuine reflective quotient requires a declared boundary identification such as

$$(\rho_{\max}, \theta, \kappa, o) \sim (\rho_{\min}, \pi - \theta, \kappa \oplus 1, -o).$$

The angular derivative is -1 . That is the orientation-reversing feature. Merely adding π to θ is a rotation and remains a different profile. [\[S3\]](#)

7.4 Discrete angle maps

For a 256-bin absolute angle code,

$$q' = (128 - q) \bmod 256.$$

For a local delta code whose chart origin is transformed separately,

$$\Delta q' = (-\Delta q) \bmod 256.$$

Both maps are exact involutions in the finite code space.

8 Canonical Glyph SDFs

8.1 Canonical profile

The reference glyph profile is `ugts.canonical.capsule-glyph.v1`. A glyph is a finite segment skeleton S_U with stroke radius r_s . Its field is

$$G_U(p) = \min_{e \in S_U} d(p, e) - r_s.$$

This is an exact Euclidean SDF to the finite union of equal-radius segment capsules that defines the canonical profile. The open ellipse is approximated by a fixed, versioned polyline; the exactness claim applies to that capsule union, not to an ideal analytic ellipse or to a third-party font.

8.2 Mirror law

For direct/converse relation pairs:

$$G_{U^{\sim}}(x, y) = G_U(-x, y).$$

The same reflection that changes visible port direction also matches the operand-converse map.

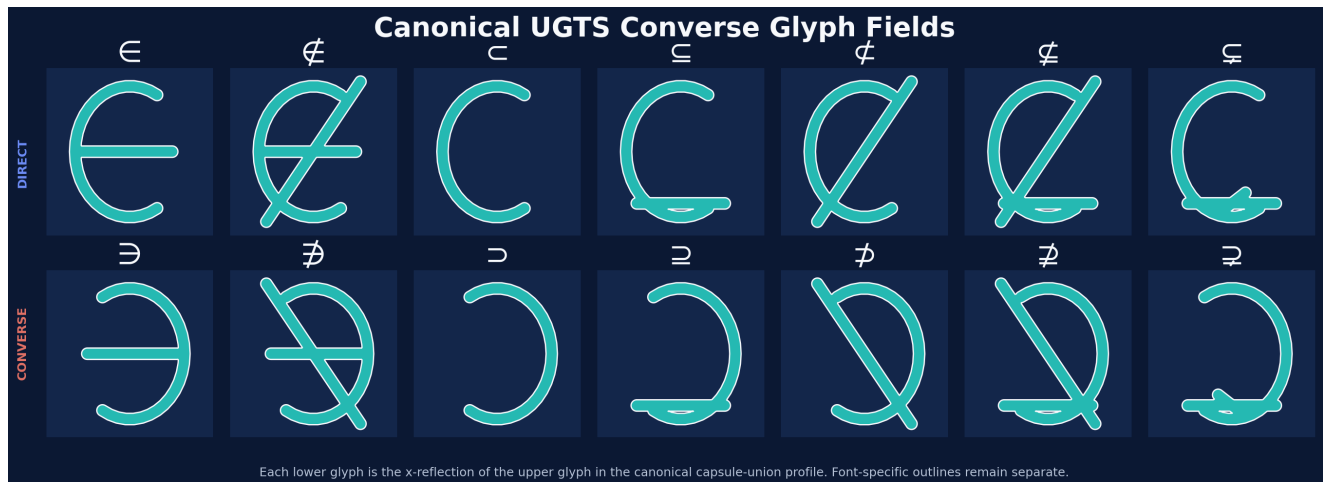


Figure 6: Fourteen shipped converse glyph fields. The lower row is the exact x-reflection of the upper row in the canonical profile.

8.3 Font-exact mode

A font profile may provide measured or extracted outlines for each literal. In that mode, font, size, style, hinting, antialias threshold, fill/stroke and connectivity are part of the record. Semantic converse remains exact, while geometric mirror equality becomes profile-bound. This directly preserves BEA's boundary/equivalence separation. [\[S2\]](#)

BOUNDARY

A Unicode codepoint fixes a character identity, not a unique visual outline. The canonical UGTS glyph is one deliberate, auditable geometry profile and never masquerades as every font.

9 Cold Atlas, Hot Codebook and LUT

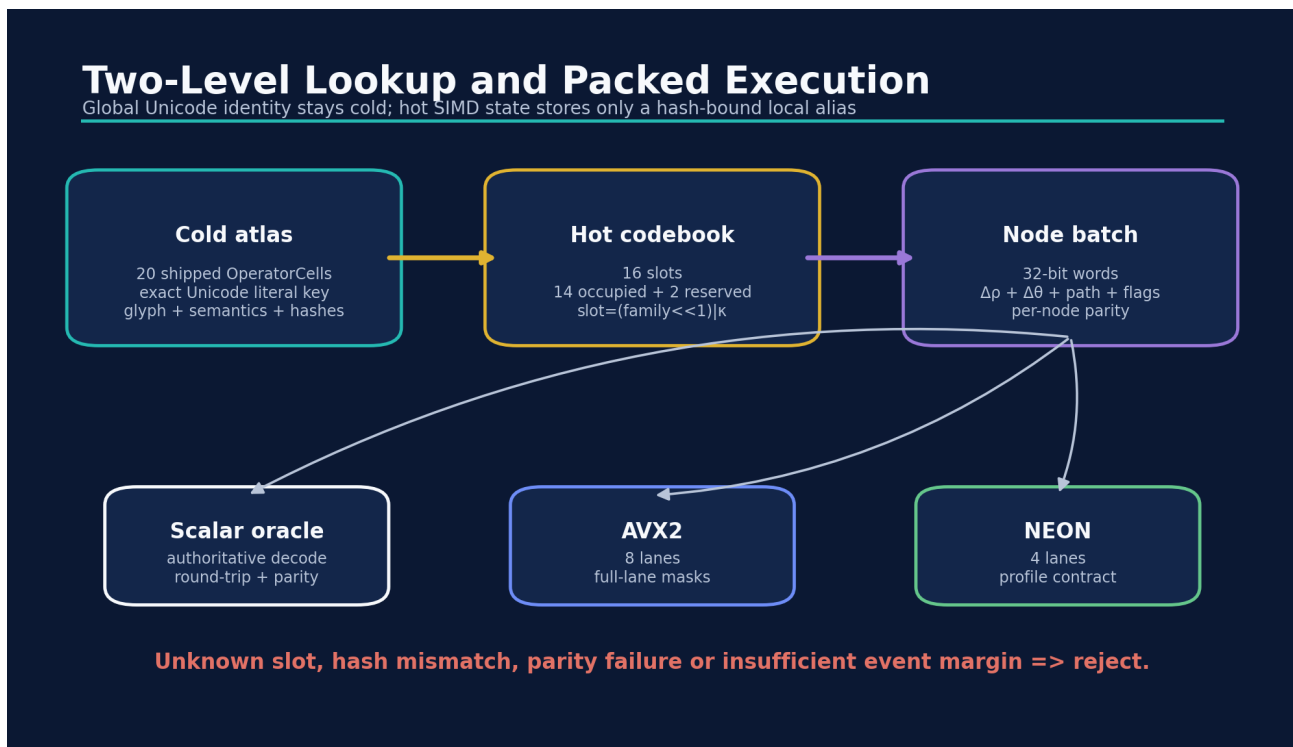


Figure 7: The local opcode is a cache alias bound to the exact global atlas.

9.1 Cold atlas

The cold atlas is keyed by exact literal Unicode sequence and contains all 20 OperatorCells. Its 5.0 hash is:

```
d4745a198469ff97665825f53d88cdbf8bad27687a3021c89177eedb83a55e91
```

It is the authority for Unicode identity, typed meaning, glyph profile, converse mapping and exactness metadata.

9.2 Hot codebook

The shipped `ugts.set-core-16.v1` codebook contains exactly 16 slots. Slots 0–13 map seven converse families; 14–15 are reserved and reject. Its hash is:

```
3a279c68331d697f31e223b71eea898a4239d4da989b6f6d5c1dfd0dfcfb009
```

Slot	Family	κ	Literal	Operator ID
0	0	0	∈	set.membership.element-of
1	0	1	⊃	set.membership.contains-as-member
2	1	0	∉	set.membership.not-element-of
3	1	1	⊄	set.membership.does-not-contain-as-member
4	2	0	⊂	set.inclusion.proper-subset
5	2	1	⊃	set.inclusion.proper-superset

Slot	Family	κ	Literal	Operator ID
6	3	0	$\subseteq$	set.inclusion.subset-or-equal
7	3	1	$\supseteq$	set.inclusion.superset-or-equal
8	4	0	$\subsetneq$	set.inclusion.not-proper-subset
9	4	1	$\supsetneq$	set.inclusion.not-proper-superset
10	5	0	$\not\subseteq$	set.inclusion.not-subset-or-equal
11	5	1	$\not\supseteq$	set.inclusion.not-superset-or-equal
12	6	0	$\subsetneqq$	set.inclusion.subset-with-not-equal
13	6	1	$\supsetneqq$	set.inclusion.superset-with-not-equal
14	-	-	reserved	-
15	-	-	reserved	-

9.3 Literal-to-LUT path

A lookup proceeds as follows:

1. Normalize and resolve the exact Unicode literal in the cold atlas.
2. Verify the cell hash and atlas hash.
3. Select or build a hot codebook whose entries commit to those hashes.
4. Store only the local four-bit slot in packed nodes.
5. Decode the slot through the codebook before semantic or glyph evaluation.

Unknown code, codebook mismatch or atlas mismatch is a hard failure.

10 Packed Log-Polar Node 32

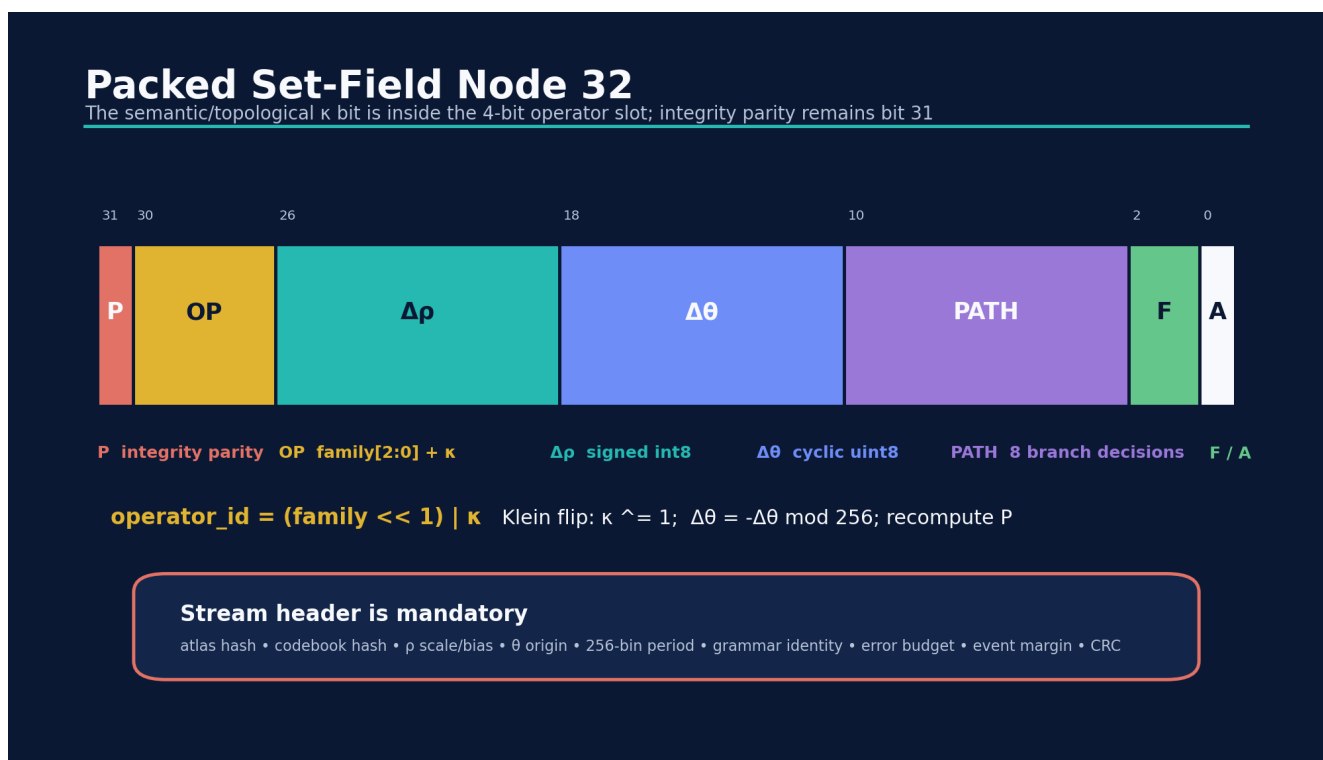


Figure 8: Corrected bit layout with separate integrity and topological roles.

10.1 Bit layout

Bits	Width	Field	Contract
31	1	integrity parity	parity of payload bits 0–30; not κ
30–27	4	operator slot	three-bit family plus one-bit κ
26–19	8	$\Delta\rho$	signed int8; scale/bias supplied by header
18–11	8	$\Delta\theta$	cyclic uint8; 256-bin period
10–3	8	grammar path	bounded eight-decision branch fragment
2–1	2	local flags	profile-defined status; narrow meanings only
0	1	active	explicitly decoded and tested

10.2 Packing formula

Let $s = (f, \kappa, \Delta\rho, \Delta\theta, p, g, a)$. The payload is

$$\text{payload} = ((2f + \kappa) \ll 27) \mid ((\Delta\rho \bmod 256) \ll 19) \\ \mid (\Delta\theta \ll 11) \mid (p \ll 3) \mid (g \ll 1) \mid a.$$

Bit 31 equals the population parity of bits 0–30, giving even parity over the full word.

10.3 Two parities, two roles

- κ is the semantic/topological direct-versus-converse selector.
- bit 31 is an integrity precheck.

A legitimate $\epsilon \rightarrow \exists$ transition toggles κ , then recomputes the integrity parity. The two roles can never share one bit.

10.4 Mandatory stream header

The 32-bit word is not self-describing. The stream header carries:

- schema, atlas and codebook IDs and hashes;
- ρ origin, scale, bias and admissible range;
- θ origin, period and rounding rule;
- reflective wrap profile;
- grammar definition and path-prefix context;
- capability and exactness policy;
- error budget and event-margin rule;
- block CRC and optional outer content hash.

The reference UG5N framing uses a canonical JSON header, little-endian 32-bit words and CRC32 over prefix, header and node payload.

10.5 Quantization

A full circle with 256 bins has bin width

$$\frac{360^\circ}{256} = 1.40625^\circ.$$

With round-to-nearest, maximum half-bin angular error is

$$0.703125^\circ.$$

Radial error depends on the declared $\Delta\rho$ scale. Packed evaluation may be promoted only when total numeric error remains below the relevant event margin.

10.6 Bounded grammar path

Eight path bits encode eight binary decisions or one fragment of a longer externally stored address. They do not encode an unbounded tree, infinite detail or a complete world.

11 Scalar and SIMD Execution Profiles

11.1 Scalar oracle

The dependency-free Python and C++20 scalar codecs are the conformance authority. Every backend must match:

- field extraction and signed $\Delta\rho$ decoding;
- active and local flags;
- operator family and κ ;
- parity verification;
- Klein flip and round trip;
- codebook resolution;
- error and capability status.

11.2 AVX2 eight-lane extraction

The optional header `native/ugts5_avx2.hpp` loads eight words and extracts fields with shifts and masks. It corrects two common source errors:

1. reading bit 31 is not parity verification; a tested popcount/XOR fold or scalar check is still required;
2. a 0/1 selector must be expanded to `0x00000000/0xffffffff` before lane-wise selection.

The reference mask expansion is conceptually

$$M = 0 - \kappa,$$

so each 32-bit lane becomes all-zero or all-one.

11.3 ARM NEON profile

The NEON contract mirrors the AVX2 layout in four lanes and uses full-lane masks before `vbslq_u32`. It is specified but not measured in this package.

11.4 Branch strategy

Evaluate-both-and-select can help when both branches are cheap and coherence is poor. It can waste work when either branch is expensive. A backend may choose among scalar branching, active-lane compaction, dual evaluation or measured automatic selection. No universal speed claim is promoted.

11.5 Benchmark boundary

Cycle counts, cache residency, SIMD utilization, latency and energy require a named processor, compiler flags, workload, precision, equal-error baseline and captured measurement. The package proves codec conformance, not target performance. [\[S4\]](#)

12 Exactness, Proof and Certification

12.1 Point query versus global relation

A point membership query can be a local field comparison. Subset, equality and emptiness are global relations:

$$A \subseteq B \iff \forall x \in X : x \in A \Rightarrow x \in B.$$

No packed local lookup turns the quantifier into universal constant-time proof.

12.2 Certificate object

A version-5 relation certificate contains:

$$C = (U, h_U, \text{profile}, \text{operands}, \text{scope}, \text{capability}, I, e, m, r, N),$$

where I is the result or interval, e numeric error, m event margin, r reason/status and N explicit non-claims.

12.3 Finite exact proof

For a declared finite universe, the runtime can exhaustively check membership, subset, proper subset, equality and emptiness. The universe ordering is part of the proof input. Changing the universe invalidates the certificate.

12.4 Continuous proof

A continuous-domain claim requires at least one of:

- an analytic proof;
- interval arithmetic covering the full domain;
- a certified spatial subdivision with a proven residual bound;
- a theorem-specific decision procedure;
- another explicitly verified completeness argument.

A sampled LUT remains sampled evidence unless the certificate closes the between-sample gap.

12.5 Integrity ladder

Layer	Role
Per-node parity	fast odd-bit corruption precheck
Stream CRC32	accidental block corruption detection
Operator-cell hash	content identity of one definition
Atlas/codebook hash	exact lookup-context binding
Pre/post state hash	authoritative transition identity
Lineage/replay	ordered explanation of how state was reached

No single layer substitutes for the others.

13 Integration with UGTS State Authority

13.1 Pure evaluation is upstream

A Unicode cell may answer a set relation, return a glyph field value, transform a signed set field, select a converse, decode a packed node or issue an exactness certificate. None of those operations changes authoritative world state by itself.

13.2 Canonical handoff

A state-changing application uses:

```
operator result + support evidence
-> compatibility
-> guard / error / uncertainty
-> verified proposal
-> deterministic conflict resolution
-> atomic commit
-> pre/post hashes
-> lineage + replay
-> optional glyph, display, storage or device projection
```

13.3 Example: membership-triggered transition

Suppose a rule proposes opening a gate when token x belongs to set A . The OperatorCell answers $x \in A$. The application must still prove that the rule is supported at this gate, the token and gate are compatible, timing and uncertainty guards pass, and the resulting patch preserves required invariants. Only then is the gate state committed.

13.4 Example: reflective route transition

A route crossing a declared radial boundary may apply $\mathcal{K}$. The local operator and orientation flip, but the transition is committed only after the boundary guard and topology profile verify. Winding and the selected quotient remain in lineage so repeated arrivals at the same coordinates are distinguishable.

14 Reference Implementation and Validation

14.1 Package map

Path	Contents
report/	PDF report and editable XeLaTeX source
src/ugts5/	canonical hashing, atlas, set fields, glyph SDF, Klein transform, packing, stream and CLI
spec/	schemas, atlas, codebook, release record, source register, mechanisms and claims
native/	C++20 scalar codec, AVX2 extraction header and NEON contract
examples/	operator, field and packed-stream demonstrations
tests/	83 standard-library Python conformance tests
validation/	captured outputs, schema results, native results, hashes and manifests
figures/	report diagrams and glyph-field plate
dist/	built Python wheel

14.2 Validation result



Figure 9: Exact captured package evidence.

Check	Scope	Result
Python conformance	83 tests	PASS
JSON Schema	3 instances	0 errors
Native scalar codec	C++20	PASS
Operator atlas	20 cells	all content hashes verified
Hot codebook	16 slots	14 assigned / 2 reserved
Mechanism delta	64 rows	USF001-USF064 / M886-M949
Claims ledger	24 rows	admit / bound / reject recorded
Wheel	py3-none-any	built and clean-installed

14.3 What the tests establish

The tests establish deterministic canonical JSON, content-hash verification, Unicode scalar/UTF-8 round trip, signed-field truth tables, finite exact set relations, glyph reflection, local Klein involution, discrete angle involution, packed round trip, active-bit decoding, integrity parity behavior, stream CRC, codebook rejection, schema validity and native scalar parity with the Python layout.

14.4 What the tests do not establish

The tests are not independent scientific validation. They do not prove every font mirrors converse glyphs, every set has a metric SDF, continuous subset claims are solved, AVX2 or NEON are faster, physical hardware is low-power, or the formalism is complete for all Unicode mathematics.

14.5 Reproduction

```
python tools/generate_specs.py
python examples/packed_stream_demo.py
python -m unittest discover -s tests -v
python tools/validate_schemas.py
c++ -std=c++20 -O2 -I native native/test_packed_node.cpp -o /tmp/ugts5_native
/tmp/ugts5_native
python -m ugts5 verify --package-root .
```

15 Claims Ledger Summary

The complete 24-row ledger is in Appendix D and in `spec/claims_ledger.json/csv`. The governing decisions are:

Claim	Disposition	Reason
Unicode can address shape and functionality	ADMIT	One typed cell co-addresses literal, glyph, evaluator and field action.
Geometry and semantics are identical	REJECT	They remain distinct domains connected by explicit maps.
Set theory can use signed set fields	ADMIT WITH CAPABILITY	Exact SDF is one class; finite, metric, residual and oracle fields remain distinct.
Converse pairs share a semantic kernel	ADMIT	Typed port normalization makes the reuse exact.
Klein flip is just a bit shift	CORRECT	It selects an involution that swaps operands and reflects the chart.
One bit can be integrity and topology	REJECT	κ and parity are separate fields.
Four-bit operator ID is global	REJECT	It is a hash-bound local codebook alias.
Packed node is complete state	REJECT	Definitions, header, uncertainty, lineage and hashes remain external.
Composed fields remain exact SDFs	REJECT	Sign is preserved; exact distance may be lost.
SIMD is automatically faster	REJECT	A named target and equal-error benchmark are required.
Deterministic parity is true randomness	REJECT	Determinism supplies repeatability, not entropy.
This atlas enumerates all mathematics	REJECT	The shipped atlas is bounded and the schema is extensible.

16 Acceptance, Kill Criteria and Final Definition

16.1 Acceptance gates

A version-5 use is accepted only when:

1. the exact Unicode scalar sequence resolves;
2. the OperatorCell and atlas hashes verify;
3. the operand types and surface-to-canonical port map validate;
4. the field capability is no stronger than the evidence;
5. the converse and glyph profile are declared;
6. the codebook and stream header match;
7. per-node parity and stream CRC pass;
8. quantization error stays below the declared event margin;
9. state mutation proceeds through the canonical authority chain;
10. lineage records accepted transitions and relevant wraps.

16.2 Kill criteria

Reject or narrow the use when:

- Unicode identity is replaced by an undocumented local opcode;
- font geometry is treated as profile-free;
- semantic zero is called glyph-SDF zero;
- a sampled LUT is presented as a continuous-domain proof;
- κ is overloaded as integrity, randomness, uncertainty or complete state;
- a half-turn is mislabeled as reflective Klein gluing;
- an unknown codebook slot aliases an existing operator;
- min/max field composition silently retains exact-SDF status;
- backend output differs from the scalar oracle;
- performance or compression claims omit a named target, workload and equal-error baseline;
- operator evaluation bypasses support, compatibility, guard, commit or lineage.

16.3 Final upgraded definition

FORMAL DEFINITION

UGTS Foundation 5.0 is a literal, content-addressed Unicode operator substrate in which set relations and set algebra may be evaluated through typed signed set fields; canonical glyph shapes are executable SDF definitions; converse surface spellings share canonical semantic kernels; reflective Klein transitions reverse typed ports, glyph orientation and chart orientation through an explicit involution; and compact log-polar execution records remain hash-bound, error-bounded and downstream of authoritative definitions.

MAJOR-VERSION CONCLUSION

A Unicode operator is no longer merely text and no longer merely an icon. It is a content-addressed, executable geometric rule cell whose literal codepoint resolves both its canonical shape and its typed operation. A Klein-converse parity flip mirrors the shape, reverses the typed ports and selects the converse Unicode spelling while preserving the mathematical truth value.

EVIDENCE BOUNDARY

The word *parity* in the conclusion names the direct/converse κ selector as a two-state orientation parity. It does not merge with bit-31 integrity parity, random entropy, uncertainty, identity or full state. The package and schemas keep those roles separate.

A Complete Shipped Unicode Operator Atlas

Lit.	Scalar	Fam.	κ	Conv.	Kernel	Stable operator ID
$\in$	U+2208	0	0	$\ni$	membership	set.membership.element-of
$\ni$	U+220B	0	1	$\in$	membership	set.membership.contains-as-member
$\notin$	U+2209	1	0	$\nexists$	nonmembership	set.membership.not-element-of
$\nexists$	U+220C	1	1	$\notin$	nonmembership	set.membership.does-not-contain-as-member
$\subset$	U+2282	2	0	$\supset$	proper subset	set.inclusion.proper-subset
$\supset$	U+2283	2	1	$\subset$	proper subset	set.inclusion.proper-superset
$\subseteq$	U+2286	3	0	$\supseteq$	subset or equal	set.inclusion.subset-or-equal
$\supseteq$	U+2287	3	1	$\subseteq$	subset or equal	set.inclusion.superset-or-equal
$\subsetneq$	U+2284	4	0	$\supsetneq$	not proper subset	set.inclusion.not-proper-subset
$\supsetneq$	U+2285	4	1	$\subsetneq$	not proper subset	set.inclusion.not-proper-superset
$\not\subseteq$	U+2288	5	0	$\not\supseteq$	not subset or equal	set.inclusion.not-subset-or-equal
$\not\supseteq$	U+2289	5	1	$\not\subseteq$	not subset or equal	set.inclusion.not-superset-or-equal
$\subsetneqq$	U+228A	6	0	$\supsetneqq$	proper subset variant	set.inclusion.subset-with-not-equal
$\supsetneqq$	U+228B	6	1	$\subsetneqq$	proper subset variant	set.inclusion.superset-with-not-equal
$\cup$	U+222A	-	-	-	union	set.algebra.union
$\cap$	U+2229	-	-	-	intersection	set.algebra.intersection
$\setminus$	U+2216	-	-	-	difference	set.algebra.difference
$\complement$	U+2201	-	-	-	complement	set.algebra.complement
$\emptyset$	U+2205	-	-	-	empty set	set.literal.empty
$=$	U+003D	-	-	-	set equality	set.relation.equality

B Hot Codebook Set-Core-16

The table in Section 9 is normative. Slots 14 and 15 are reserved and must reject unless a different versioned codebook is selected in the stream header. The four-bit value is never globally meaningful without the codebook hash.

C Complete Mechanism Delta USF001-USF064

ID	Domain	Name	Definition	Validation
USF001 / M886	release	Release identity	Component-scoped major release identity with legacy UGTS-KC 5.0 alias.	implemented-and-tested
USF002 / M887	source	Source register	Content-hashed source and parent register without raw-source redistribution.	implemented-and-tested
USF003 / M888	referential	OperatorCell	One content-addressed record co-addresses literal, glyph, semantics and field action.	implemented-and-tested
USF004 / M889	unicode	Unicode scalar key	Exact Unicode scalar sequence is a canonical cold-atlas lookup key.	implemented-and-tested

ID	Domain	Name	Definition	Validation
USF005 / M890	unicode	UTF-8 literal key	UTF-8 spelling is recorded and round-tripped without ASCII aliasing.	implemented-and-tested
USF006 / M891	unicode	NFC normalization profile	NFC is explicit; compatibility normalization is not silently applied.	implemented-and-tested
USF007 / M892	typing	Typed signature	Every operator declares domain, codomain, arity and type parameters.	implemented-and-tested
USF008 / M893	typing	Canonical argument order	One semantic port order is authoritative for each converse family.	implemented-and-tested
USF009 / M894	typing	Surface argument map	Surface operands map explicitly to canonical semantic ports.	implemented-and-tested
USF010 / M895	semantics	Semantic evaluator	Literal dispatch selects a bounded typed semantic kernel.	implemented-and-tested
USF011 / M896	geometry	Canonical glyph SDF	A versioned capsule-union field represents the literal operator shape.	implemented-and-tested
USF012 / M897	field	Set-field action	Operator records declare predicates or sign-compatible field transforms.	implemented-and-tested
USF013 / M898	relation	Converse link	Direct and converse cells reference each other by stable operator ID.	implemented-and-tested
USF014 / M899	integrity	Operator content hash	Every cell carries a canonical SHA-256 content address.	implemented-and-tested
USF015 / M900	provenance	Provenance class	Source motif, engineering normalization and measured evidence stay distinct.	implemented-and-tested
USF016 / M901	exactness	Capability class	Exact SDF, metric field, bound, residual, membership field and oracle remain distinct.	implemented-and-tested
USF017 / M902	field	Signed set field	$\phi_A = d(x,A) - d(x,X \setminus A)$ under a declared metric profile.	implemented-and-tested
USF018 / M903	field	Characteristic fall-back	Finite or non-metric sets can use a signed membership field.	implemented-and-tested
USF019 / M904	relation	Membership predicate	$x \in A$ iff $\phi_A(x) \leq 0$ under the selected boundary policy.	implemented-and-tested
USF020 / M905	relation	Nonmembership predicate	$x \notin A$ is the typed negation of membership.	implemented-and-tested
USF021 / M906	relation	Subset predicate	A subseteq B is a quantified emptiness claim over $A \setminus B$.	implemented-and-tested
USF022 / M907	relation	Proper subset predicate	Proper inclusion adds non-equality to subset.	implemented-and-tested
USF023 / M908	relation	Set equality	$A=B$ is mutual inclusion under a common universe/profile.	implemented-and-tested
USF024 / M909	relation	Empty-set predicate	Emptiness is exact only when the domain proof is complete.	implemented-and-tested
USF025 / M910	field	Union field	$\min(\phi_A, \phi_B)$ preserves the union sign.	implemented-and-tested
USF026 / M911	field	Intersection field	$\max(\phi_A, \phi_B)$ preserves the intersection sign.	implemented-and-tested
USF027 / M912	field	Complement field	$-\phi_A$ reverses inside and outside.	implemented-and-tested
USF028 / M913	field	Difference field	$\max(\phi_A, -\phi_B)$ preserves the sign of $A \setminus B$.	implemented-and-tested
USF029 / M914	field	Symmetric-difference field	A sign-correct max/min formula represents $A \triangle B$.	implemented-and-tested
USF030 / M915	exactness	Field capability downgrade	Composed fields never inherit exact-distance status without proof.	implemented-and-tested
USF031 / M916	proof	Finite-universe proof	Exhaustive finite checks may issue exact relation certificates.	implemented-and-tested
USF032 / M917	proof	Interval/analytic proof	Continuous global claims require analytic, interval or certified-cover evidence.	implemented-and-tested
USF033 / M918	proof	Sample-only non-proof	A sampled LUT result is bounded evidence, not universal inclusion proof.	implemented-and-tested
USF034 / M919	topology	Klein-converse transform	One transform reflects chart, swaps ports and toggles converse state.	implemented-and-tested
USF035 / M920	topology	Operator converse	U maps to U-smile while preserving truth after operand exchange.	implemented-and-tested
USF036 / M921	topology	Operand swap	Typed endpoints reverse rather than being reinterpreted in place.	implemented-and-tested
USF037 / M922	geometry	Glyph reflection	Canonical converse glyph is the x-reflection of the direct glyph.	implemented-and-tested
USF038 / M923	topology	Angular reflection	θ maps to $\pi - \theta$ on the reflective profile.	implemented-and-tested
USF039 / M924	topology	Orientation selector	κ is a semantic/topological selector and not an integrity bit.	implemented-and-tested

ID	Domain	Name	Definition	Validation
USF040 / M925	topology	Involution law	Applying the Klein-converse transform twice returns the original state.	implemented-and-tested
USF041 / M926	topology	Reflective quotient	Boundary gluing declares the actual orientation-reversing quotient.	implemented-and-tested
USF042 / M927	topology	Half-turn distinction	A pi rotation remains distinct from a reflective Klein gluing.	implemented-and-tested
USF043 / M928	atlas	Cold Unicode atlas	The full literal record remains content-addressed outside hot SIMD state.	implemented-and-tested
USF044 / M929	atlas	Hot 16-slot codebook	A batch-local codebook maps four-bit slots to exact atlas cells.	implemented-and-tested
USF045 / M930	atlas	Reserved-slot rejection	Unassigned local codes reject rather than aliasing silently.	implemented-and-tested
USF046 / M931	integrity	Codebook content hash	Hot mappings are bound to the exact cold atlas.	implemented-and-tested
USF047 / M932	packing	Packed Set-Field Node 32	A corrected 32-bit execution record carries local deltas and flags.	implemented-and-tested
USF048 / M933	integrity	Integrity parity	Bit 31 checks payload parity and remains separate from kappa.	implemented-and-tested
USF049 / M934	packing	Family/kappa extraction	operator_id=(family<<1) kappa enables converse-pair locality.	implemented-and-tested
USF050 / M935	packing	Signed log-radius delta	An explicit int8 delta requires a stream scale/bias contract.	implemented-and-tested
USF051 / M936	packing	Cyclic angle delta	An 8-bit angular delta has a declared 256-bin period.	implemented-and-tested
USF052 / M937	grammar	Grammar path fragment	Eight bits represent a bounded branch fragment, not an unbounded tree.	implemented-and-tested
USF053 / M938	packing	Local field flags	Two bits carry profile-defined local status with declared meaning.	implemented-and-tested
USF054 / M939	packing	Active bit	Bit zero is decoded and tested rather than silently dropped.	implemented-and-tested
USF055 / M940	stream	Stream header	Atlas, codebook, chart, quantization and error contracts travel with nodes.	implemented-and-tested
USF056 / M941	integrity	Block CRC	Stream-level CRC supplements per-node parity and content hashes.	implemented-and-tested
USF057 / M942	runtime	Scalar oracle	Dependency-free scalar decoding is the conformance authority.	implemented-and-tested
USF058 / M943	simd	Branchless mask expansion	Boolean selectors expand to full-lane masks before blending.	implemented-and-tested
USF059 / M944	simd	AVX2 profile	Eight-lane extraction is specified behind the scalar oracle.	specified-contract
USF060 / M945	simd	NEON profile	Four-lane ARM decoding is a specified backend target.	specified-contract
USF061 / M946	quantization	Quantization certificate	Angular/radial errors are calculated from the declared profile.	implemented-and-tested
USF062 / M947	guard	Event-margin guard	Packed evaluation is accepted only below the relevant event margin.	implemented-and-tested
USF063 / M948	authority	UGTS authority handoff	Pure operator evaluation cannot bypass support, compatibility and commit.	implemented-and-tested
USF064 / M949	validation	Validation and kill criteria	Claims fail closed on hash, type, parity, exactness or evidence violations.	implemented-and-tested

D Complete Claims Ledger

ID	Claim	Disposition	Reason
USF-C01	Unicode literal can key shape and functionality	ADMIT	The same content-addressed cell co-addresses literal, glyph, evaluator and field action.
USF-C02	Geometry and semantics are identical	REJECT	They remain typed faces of one record; equality requires an explicit map/certificate.
USF-C03	Set theory can use signed set fields	ADMIT WITH CAPABILITY	Exact SDF is one capability; finite, metric, residual and oracle profiles are distinct.
USF-C04	Membership is a local sign query	ADMIT	Under a declared set-field profile, x in A iff $\phi_A(x) \leq 0$.
USF-C05	Every subset claim is $O(1)$	REJECT	Global inclusion is quantified and needs finite exhaustion or a continuous-domain proof.

ID	Claim	Disposition	Reason
USF-C06	Converse pairs share one kernel	ADMIT	Surface operand maps and typed endpoints make the reuse exact.
USF-C07	Klein flip is merely a bit shift	CORRECT	The bit selects an involution that also swaps operands and reflects the chart.
USF-C08	Klein-converse preserves truth	ADMIT	$E_converse(b,a)=E_direct(a,b)$ for registered relation pairs.
USF-C09	A half-turn is a Klein reflection	REJECT	Rotation and orientation-reversing gluing remain distinct profiles.
USF-C10	One bit can be both integrity and topology	REJECT	Integrity parity and kappa are separate logical and physical fields.
USF-C11	Canonical glyph converse is a mirror	ADMIT PROFILE BOUND	True for the UGTS canonical capsule profile; font-exact profiles may differ.
USF-C12	Unicode codepoint alone fixes font appearance	REJECT	Font, size, stroke, threshold and rendering policy remain profile data.
USF-C13	Four-bit operator ID is globally unique	REJECT	It is only a hot codebook slot bound to an atlas hash.
USF-C14	Packed node is complete world state	REJECT	Definitions, header, lineage, uncertainty and hashes remain external.
USF-C15	One parity bit detects all corruption	REJECT	It detects odd payload bit flips; CRC/hash layers cover stronger integrity roles.
USF-C16	32-bit record reduces traffic	ADMIT CONDITIONAL	Only under a measured workload and an error/equivalence contract.
USF-C17	Eight-bit angle gives 1.40625 degree bins	ADMIT	360/256 with round-to-nearest maximum half-bin error 0.703125 degrees.
USF-C18	Composed min/max fields remain exact SDFs	REJECT	Sign is preserved, exact-distance status may be lost.
USF-C19	Sampled LUT proves continuous emptiness	REJECT	Sampling can miss features between samples unless a certificate closes the gap.
USF-C20	SIMD branchless selection is always faster	REJECT	Backend choice depends on branch cost, coherence and target measurement.
USF-C21	AVX2 eight-lane extraction is feasible	ADMIT AS PROFILE	Correct extraction and full-lane masks are supplied; target throughput remains benchmarked evidence.
USF-C22	Deterministic parity is true randomness	REJECT	Deterministic bits provide repeatable selection or checks, not entropy.
USF-C23	16 KB stores arbitrary worlds	REJECT	Small atlases and repetitive grammars may fit; arbitrary information does not.
USF-C24	This release enumerates all Unicode mathematics	REJECT	The schema is extensible; the shipped atlas is a bounded set-theory foundation.

E Source Hashes and Package Reproduction

E.1 Full source hash register

```

S0 6149e3d398c22b53d0664ba2f624c046d6260506f9ac009dfb2b2572afb3d18c
S1 e21433cd8378368dafa4e278d3c9c364725e45ae7fd75b743328bd2d6003a46
S2 ac857c9c01f5f4e93edbc8657d2d549a085d939dd1cedce3829ab62bc1dca096
S3 6987490a0a7b7ed76b0091057a2e69ab831fee786434a0b4219782923f5ea580
S4 3a6b6c57118ca4e2d7d7e577508c3477a095d133791f3708cb56a054285225d2
S5 413ef9568330e253ebb8e42cbc756257f3c138485f2d9697de48324b92f28efe
S6 c0ae58652436476f77be3b1f94d4b82cf66f8ba3ca5ae25ad5f6706b217ed8d4
S7 1b64495ed32a24a66f356652cb6fb1291d0727ab9bfa9174c994d757325f037b

```

E.2 Artifact identity

The final package includes validation/SHA256SUMS, a CSV manifest and a plain relative-path manifest. These identify the delivered bytes; they do not establish legal authorship or priority.

E.3 Canonical filename

UGTS__FOUNDATION__UNICODE_SET_FIELD_KLEIN_CONVERSE__v5.0.0__2026-08-29.pdf
UGTS__FOUNDATION__UNICODE_SET_FIELD_KLEIN_CONVERSE__v5.0.0__2026-08-29.zip

E.4 Release status

RELEASED AS A COMPLETE TECHNICAL FOUNDATION PACKAGE. The formal schema, reference runtime, examples, tests, native codec and evidence are delivered. Physical CPU/GPU/phone performance, independent mathematical review and font-wide geometry equivalence remain open validation activities rather than implied results.